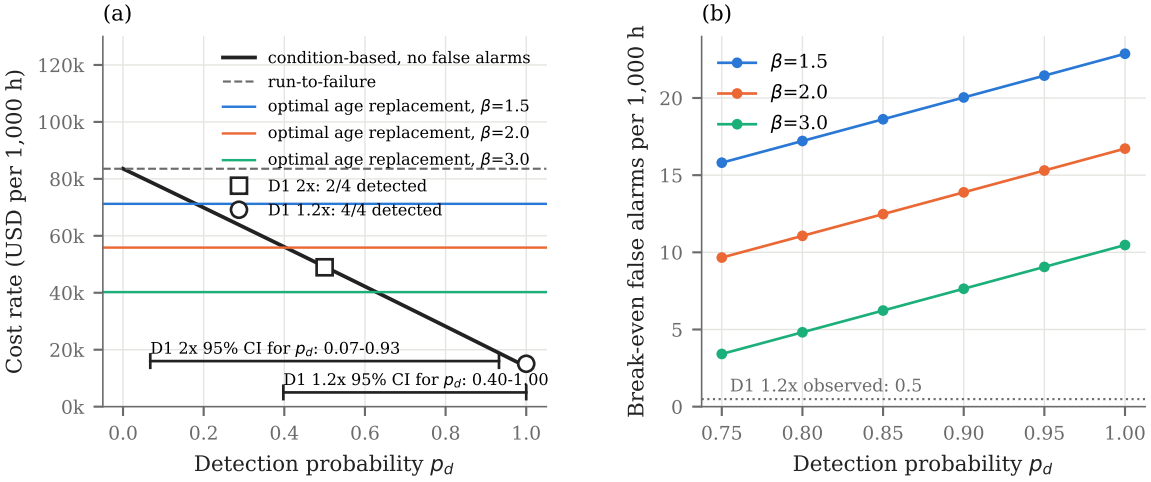
(a)
Cost rate (USD per 1,000 h)
120k
100k
80k
60k
40k
20k
0k
0.0
0.2
0.4
0.6
0.8
1.0
Detection probability p_d
condition-based, no false alarms
run-to-failure
optimal age replacement, β=1.5
optimal age replacement, β=2.0
optimal age replacement, β=3.0
D1 2x: 2/4 detected
D1 1.2x: 4/4 detected
D1 2x 95% CI for p_d: 0.07-0.93
D1 1.2x 95% CI for p_d: 0.40-1.00
(b)
Break-even false alarms per 1,000 h
20
15
10
5
0
0.75
0.80
0.85
0.90
0.95
1.00
Detection probability p_d
β=1.5
β=2.0
β=3.0
D1 1.2x observed: 0.5